MICROSOFT CHALLENGE — MULTIMEDIA PROJECT

Hand Gesture Classification on Edge Devices

Team 14 · Compact gesture classifier for real-world deployment

Presentation Agenda

01

Problem & Scoring
Definition and rules

02

Model Architecture
Dual-Stream Fusion

03

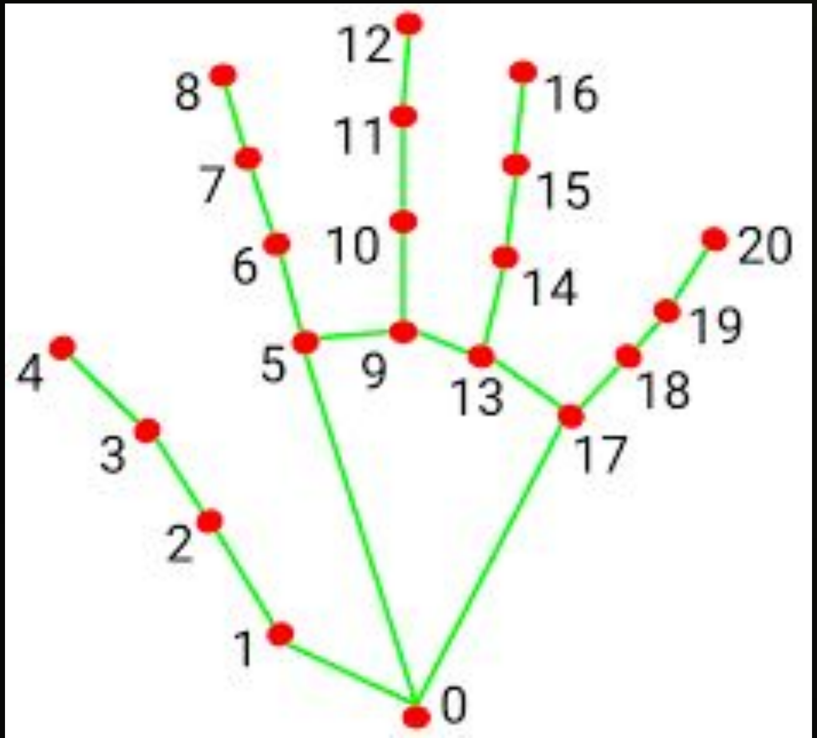
Training & Results
Pipeline and accuracy

Project Problem Definition

Input Streams

- **Cropped RGB Image:** MediaPipe pre-cropped, variable size.
- **21 Landmarks:** (x, y) relative coordinates [0, 1].

> **Constraint:** Full-frame images are strictly prohibited.



Model Scoring Rules

Criterion	Points	Formula
Model Size ≤ 10 MB	30	(10 – size_MB) × 3
Performance (HaGRIDv2)	20	+1 correct, **−2 false trigger**
Robustness (TA-shot)	40	50 N/A + 50 target images; same scoring
Presentation	30	Live demo + explanation

Key Insight: False trigger = −2 pts | Correct = +1 pt. Conservative N/A rejection is the highest-leverage optimization.

Model Architecture

Dual-Stream Fusion

Combining visual texture (CNN) with structural geometry (MLP) for robust classification.

36K

Parameters

0.142

Size (MB)

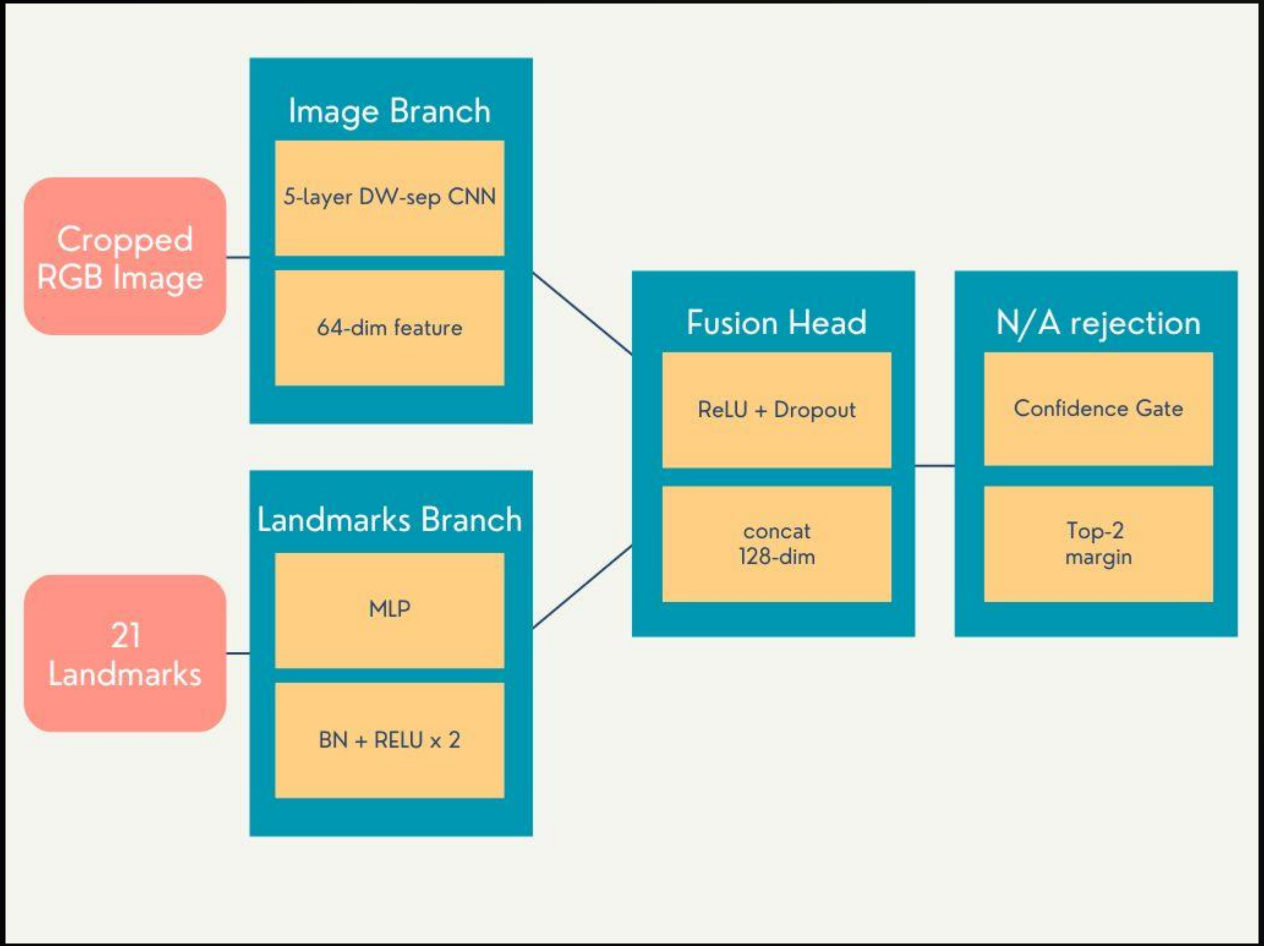


Image Branch CNN Detail

MobileNet-style Depthwise-Separable CNN

Block = DWConv → BN → ReLU → PWConv → BN → ReLU

Block	In/Out	Stride
Block 1	64→32	2
Block 2	32→16	2
Block 3	16→8	2
Block 4/5	8→8	1

Efficiency Win, Smaller and Faster

DW-sep conv uses **~8–9× fewer** operations than standard convolutions, enabling deployment on extreme edge constraints.

Landmark Branch & Fusion Head

Landmark Branch

Input: 42 floats (21 pts). Encodes **structural geometry** independent of lighting or color via a 2-layer MLP.

Fusion Head

Concatenates both features (128-dim). Uses Dropout (0.3) to prevent co-adaptation between streams.

Complementary

Texture + Structure. Neither alone achieves the same accuracy as the dual-stream approach.

N/A Rejection 2-Layer Defense

Layer 1

Confidence Threshold

Output N/A **if max(prob) < thresh**.

Handles network uncertainty.

Layer 2

Top-2 Margin Threshold

Output N/A **if 1st – 2nd < margin**.

Handles ambiguous tied classes.

Training Pipeline Overview

Stage 1

Image Pretraining

CNN warm-start (20 Epochs)

Val Acc: **86.15%**

Stage 2

Joint Training

Full Dual-Stream (30 Epochs)

Val Acc: **96.75%**

Stage 3

Calibration

Threshold Grid Search

154 Combinations

Dataset HaGRIDv2 512px

Split	N/A (0)	Target (1-5)	Total
Train	487,094	~106,000	593,703
Val	58,407	~13,500	71,982
Test	99,385	~23,000	122,780

Imbalance Alert

N/A class is **~8× larger** than any target class. Handled via Weighted Sampler and Scoring-Aware Loss.

Data Augmentation TA-Sync

Transform	Parameter
Bbox Jitter	±10% expansion
Gaussian Blur	σ 0.1–1.5
Rotation	±15°
Color Jitter	±20% intensity

Landmark Sync

Landmarks must rotate with the image:

$$x' = (x - 0.5) \cdot \cos\theta - (y - 0.5) \cdot \sin\theta + 0.5$$

Phase 1 Image Pretraining

Warm-start Strategy

- Adam Optimizer, lr = 1e-3, **20 epochs**, batch_size=256
- Cosine Annealing Scheduler LR
- **WeightedRandomSampler**: Balances 8× N/A imbalance

86%

BEST VALIDATION ACCURACY

CNN weights copied for Phase 2

Phase 2 Joint End-to-End

96.7%

BEST VALIDATION ACCURACY

Optimization Tricks

- 30 epochs, batch_size=256

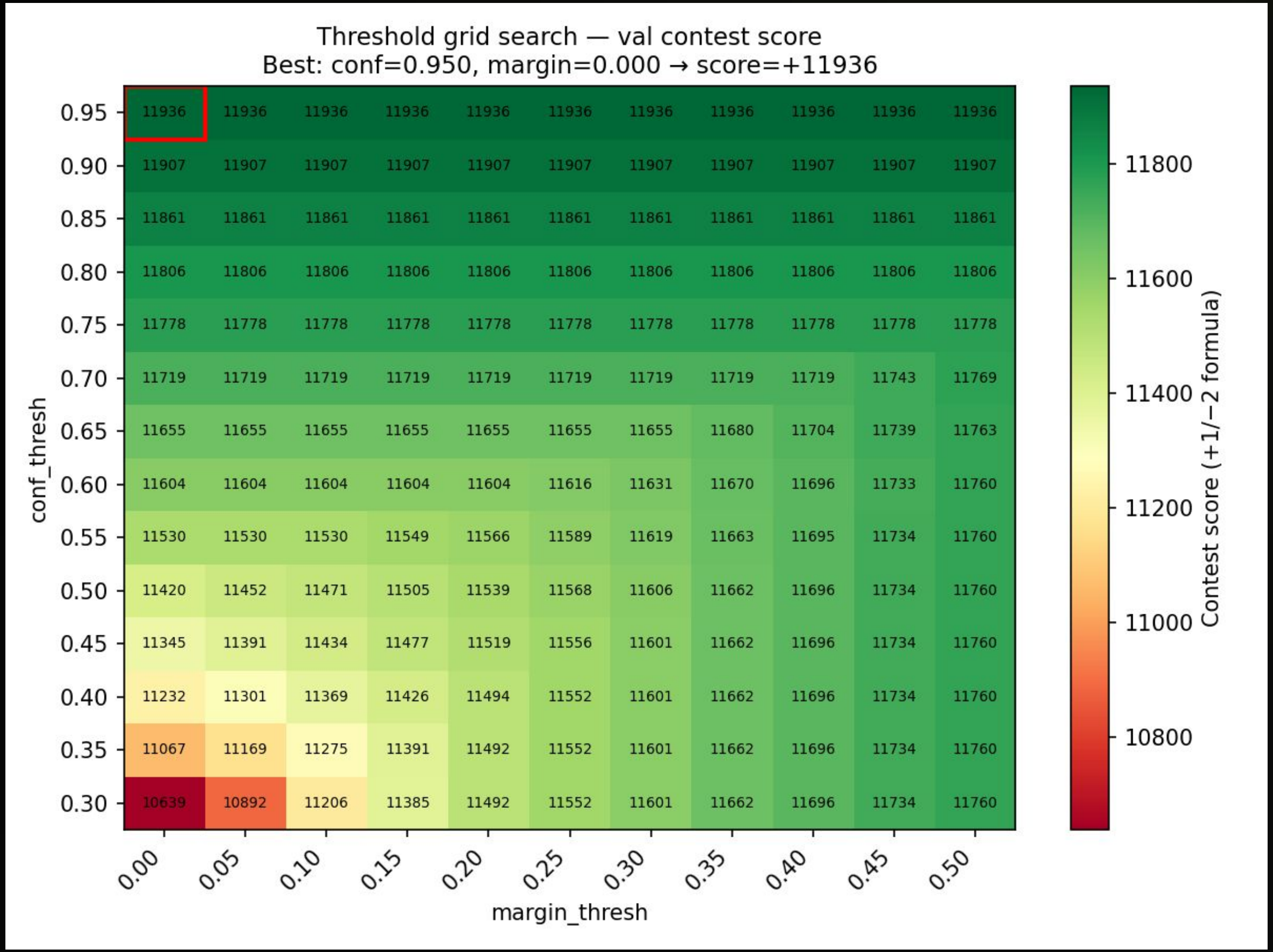
Differential Learning Rates:

- Image Branch: 1e-4
- Landmark + Fusion: 5e-4

Weighted Cross-Entropy: N/A class weight × **1.5 boost** to penalize false triggers.

Calibration Threshold Grid Search

Heatmap: Contest Score



Grid search over **154 combinations** to maximize contest score:

- conf_thresh: 0.30 to 0.95 (14 steps)
- margin_thresh: 0.00 to 0.50 (11 steps)
- **Best: conf_thresh: 0.95, margin_thresh: 0.0**

> Scoring Formula: +1 correct, −2 false trigger, 0 N/A.

Final Results Summary

We trained 2 runs

Rank	Model Size (30)	Basic Performance (20)	Real World Robustness (40)	Total Score (90)
🥇 1	29.5	19.4	32.5	81.4
🥈 2	29.5	20	31	80.5

Performance

Rank1, 2

Best Contest Score

Efficiency

0.142MB

Total FP32 Model Size

Total Score

81.4

Points (Max 90)

TEAM 14 PRESENTATION

Video Demo

Showcasing real-time recognition of target gestures and robust N/A rejection of ambiguous poses.

Youtube Link: https://youtu.be/NY15sSkPXS0?si=1D6y46Xl8HaMoeTL

▶ fist · like · ok · one · palm